

Samuele Lolli

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Education

University of Bologna

Sept 2018 – July 2022

Bachelor's Degree in Computer Science for management

- **Key subjects:** Algorithms and Data Structures, Software Engineering, Web Technologies, Databases
- **Thesis:** Machine learning for house price prediction

University of Bologna

Sept 2022 – Dec 2025

Master's Degree in Computer Science

- **Key subjects:** Data Analytics, Artificial Intelligence, Deep Learning, Scalable and Cloud Programming, Emerging Programming Paradigms, Natural Language Processing
- **Thesis:** Aggregation techniques in federated learning: heterogeneity and robustness (110 out of 110 with honors) [View on GitHub](#) 

Experience

Frontend Developer Intern

Bologna

Luna Partner SRL

Nov 2021 – Dec 2021

- Contributed to the development of features for both a web and mobile application using Angular and React

Research Intern

Bologna

Consiglio Nazionale delle ricerche (CNR) - Bologna

Ott 2025 – Dec 2025

- Create a deep learning-based model to accelerate 3D fluid simulation. The goal is to replace or complement expensive classical numerical solvers with a neural operator (FNO3d) capable of instantly predicting velocity and pressure fields given variable geometry and boundary conditions. [View on GitHub](#) 

Technologies

Programming Languages and Machine Learning tools: Python, Javascript, Java

Machine Learning Tools: Pandas, Scikit-Learn, PyTorch, Hugging Face, Spark, Flower

Web Development: React, Next, Node

Certifications

Blockchain Specialization – Buffalo University, Coursera

[View Certificate](#) 

Supervised Machine Learning – Stanford Online & Deep Learning AI, Coursera

[View Certificate](#) 

Projects

Steam Recommendation System

[View on GitHub](#) 

- Developed a recommendation system using Scala and Spark, employing TF-IDF and Cosine Similarity to deliver personalized recommendations
- Deployed on Google Cloud DataProc with a cluster comprising 7 worker nodes and 1 master node

Fine-Tuning Google's Pegasus Model

[View on GitHub](#) 

- Fine-tuning for news summarization tasks using Pytorch and Hugging Face Transformers library
- Evaluated model performance pre and post fine-tuning using ROUGE metrics

Data Analytics Techniques

[View on GitHub](#) 

- Predicted the release year of music tracks based on acoustic features
- The pipeline includes data visualization, preprocessing, model building, and training of both classical machine learning models and advanced neural networks

MilkyWay

[View on GitHub](#) 

- Developed a decentralized application for tracking the supply chain in the milk industry, improving transparency for end consumers
- Tools: Solidity, Ganache, Truffle, React.js, Node.js

Student Dropout Prediction

- Based on a real dataset provided by the University of Bologna with information on students career paths, machine learning techniques were used to predict university dropout among students.

Mobility Tracking System

- This platform monitors urban mobility in Bologna by collecting crowdsourced data from voluntary users. The system aggregates this information to estimate average travel speeds for various modes of transport while employing specific algorithms to protect user privacy without compromising service quality. Additionally, it features a front-end dashboard that allows for the visualization and filtering of the collected data by date and time.